口腔诊所管理系统｜完整逻辑 ERD •20 张表

▨ 诊所、岗位与基础资料
▨ 患者、预约与诊疗
▨ 收费明细、支付与催收
▨ 义齿委托、交付与通知

clinic｜诊所资料

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	clinic_code	VARCHAR(32)	
	name	VARCHAR(120)	
	address	VARCHAR(240)?	
	phone	VARCHAR(32)?	
	active	BOOLEAN	
	created_at	TIMESTAMP	
	updated_at	TIMESTAMP	

employee｜员工

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK	clinic_id	BIGINT	clinic.id
UK	employee_no	VARCHAR(32)	
	name	VARCHAR(80)	
	phone	VARCHAR(32)?	
	active	BOOLEAN	
	joined_on	DATE?	
	left_on	DATE?	

patient｜患者档案

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	patient_no	VARCHAR(32)	
	name	VARCHAR(80)	
	birth_date	DATE?	
	phone	VARCHAR(32)?	
	contact_name	VARCHAR(80)?	
	contact_phone	VARCHAR(32)?	
	important_note	TEXT?	
FK	created_by	BIGINT	employee.id
	created_at	TIMESTAMP	
	updated_at	TIMESTAMP	

price_item｜收费价目

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	item_code	VARCHAR(32)	
	name	VARCHAR(80)	
FK	unit_id	BIGINT	basic_option.id
	unit_price	DECIMAL(12,2)	
	active	BOOLEAN	
	updated_at	TIMESTAMP	

bill｜收费单与催收

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	bill_no	VARCHAR(32)	
FK	encounter_id	BIGINT	encounter.id
FK	issued_by	BIGINT	employee.id
	discount_amount	DECIMAL(12,2)	
	discount_reason	TEXT?	
	issued_at	TIMESTAMP	
	due_date	DATE?	
	debt_opened_at	TIMESTAMP?	
	settled_at	TIMESTAMP?	
	collection_note	TEXT?	
	stage	VARCHAR(24)	
	voided_at	TIMESTAMP?	
	void_reason	TEXT?	

product｜义齿产品

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	product_code	VARCHAR(32)	
	name	VARCHAR(80)	
	material	VARCHAR(80)	
FK	unit_id	BIGINT	basic_option.id
	specification	VARCHAR(240)?	
	requires_attachment	BOOLEAN	
	active	BOOLEAN	

dental_lab｜合作技工所

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	lab_code	VARCHAR(32)	
	name	VARCHAR(120)	
	contact	VARCHAR(80)	
	phone	VARCHAR(32)	
	address	VARCHAR(240)?	
	interface_code	VARCHAR(64)?	
	active	BOOLEAN	

clinic_room｜诊室

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK	clinic_id	BIGINT	clinic.id
UK	room_code	VARCHAR(24)	
	name	VARCHAR(60)	
	floor_note	VARCHAR(80)?	
	active	BOOLEAN	

employee_role｜员工兼岗

键	字段	SQL 类型	外键目标
PK,FK	employee_id	BIGINT	employee.id
PK,FK	role_id	BIGINT	basic_option.id
	assigned_at	TIMESTAMP	
	revoked_at	TIMESTAMP?	

appointment｜诊疗预约

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	appointment_no	VARCHAR(32)	
FK	patient_id	BIGINT	patient.id
FK	doctor_id	BIGINT	employee.id
FK	room_id	BIGINT	clinic_room.id
FK	type_id	BIGINT	basic_option.id
FK	created_by	BIGINT	employee.id
	chief_complaint	TEXT?	
	starts_at	TIMESTAMP	
	ends_at	TIMESTAMP	
	stage	VARCHAR(24)	
	created_at	TIMESTAMP	
	confirmed_at	TIMESTAMP?	
	arrived_at	TIMESTAMP?	
	completed_at	TIMESTAMP?	
	cancelled_at	TIMESTAMP?	
	cancel_reason	TEXT?	

bill_line｜收费项目明细

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK,UK*	bill_id	BIGINT	bill.id
UK*	line_no	INTEGER	
FK	price_item_id	BIGINT	price_item.id
FK	dental_order_id	BIGINT?	dental_order.id
	item_name	VARCHAR(80)	
	unit	VARCHAR(24)	
	quantity	DECIMAL(10,2)	
	unit_price	DECIMAL(12,2)	
	discount_amount	DECIMAL(12,2)	
	description	VARCHAR(240)?	

payment｜收款退款与支付尝试

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK	bill_id	BIGINT	bill.id
UK	request_no	VARCHAR(64)	
UK	external_txn_no	VARCHAR(96)?	
FK	method_id	BIGINT	basic_option.id
FK	refund_of_id	BIGINT?	payment.id
	kind	VARCHAR(24)	
	result	VARCHAR(24)	
	amount	DECIMAL(12,2)	
	requested_at	TIMESTAMP	
	occurred_at	TIMESTAMP?	
	failure_reason	TEXT?	
	checked_at	TIMESTAMP?	

dental_order｜义齿委托订单

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	order_no	VARCHAR(32)	
FK	encounter_id	BIGINT	encounter.id
FK	product_id	BIGINT	product.id
FK	lab_id	BIGINT	dental_lab.id
FK	created_by	BIGINT	employee.id
FK	color_id	BIGINT	basic_option.id
FK	box_color_id	BIGINT?	basic_option.id
FK	type_id	BIGINT	basic_option.id
FK	unit_id	BIGINT	basic_option.id
	quantity	DECIMAL(10,2)	
	repair_requirement	TEXT	
	promised_date	DATE	
	stage	VARCHAR(24)	
	dispatch_result	VARCHAR(24)	
	expected_at	TIMESTAMP?	
	created_at	TIMESTAMP	
	sent_at	TIMESTAMP?	
	accepted_at	TIMESTAMP?	
	trial_at	TIMESTAMP?	
	trial_result	VARCHAR(24)?	
	delivered_at	TIMESTAMP?	
	cancelled_at	TIMESTAMP?	
	cancel_reason	TEXT?	

order_event｜技工所回执与进展

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK,UK*	order_id	BIGINT	dental_order.id
UK*	source	VARCHAR(64)	
UK*	event_no	VARCHAR(64)	
FK	changed_by	BIGINT?	employee.id
	old_stage	VARCHAR(24)	
	new_stage	VARCHAR(24)	
	reason	TEXT?	
	external_order_no	VARCHAR(64)?	
	occurred_at	TIMESTAMP	
	received_at	TIMESTAMP	

basic_option｜分类基础选项

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK*	kind	VARCHAR(24)	
UK*	code	VARCHAR(32)	
	name	VARCHAR(80)	
	sort_order	INTEGER	
	active	BOOLEAN	
	description	VARCHAR(240)?	

appointment_event｜预约变更历史

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK	appointment_id	BIGINT	appointment.id
FK	changed_by	BIGINT	employee.id
	old_start	TIMESTAMP?	
	new_start	TIMESTAMP?	
	old_end	TIMESTAMP?	
	new_end	TIMESTAMP?	
	old_stage	VARCHAR(24)	
	new_stage	VARCHAR(24)	
	reason	TEXT?	
	occurred_at	TIMESTAMP	

encounter｜诊疗记录

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
UK	encounter_no	VARCHAR(32)	
FK	patient_id	BIGINT	patient.id
FK	clinic_id	BIGINT	clinic.id
FK	doctor_id	BIGINT	employee.id
FK,UK	appointment_id	BIGINT?	appointment.id
	occurred_at	TIMESTAMP	
	chief_complaint	TEXT	
	diagnosis	TEXT?	
	treatment_note	TEXT?	
	completed_at	TIMESTAMP?	
	created_at	TIMESTAMP	

order_tooth｜订单牙位

键	字段	SQL 类型	外键目标
PK,FK	order_id	BIGINT	dental_order.id
PK	tooth_code	VARCHAR(8)	
	repair_note	VARCHAR(240)?	

order_attachment｜修复附件

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK	order_id	BIGINT	dental_order.id
FK	type_id	BIGINT	basic_option.id
FK	uploaded_by	BIGINT	employee.id
	file_name	VARCHAR(180)	
UK	storage_key	VARCHAR(240)	
	mime_type	VARCHAR(80)	
	byte_size	BIGINT	
	checksum	VARCHAR(64)?	
	uploaded_at	TIMESTAMP	

reminder｜预约与订单通知

键	字段	SQL 类型	外键目标
PK	id	BIGINT	
FK,UK*	order_id	BIGINT?	dental_order.id
FK,UK*	appointment_id	BIGINT?	appointment.id
FK	channel_id	BIGINT	basic_option.id
FK	created_by	BIGINT	employee.id
UK*	kind	VARCHAR(24)	
UK*	scheduled_at	TIMESTAMP	
UK*	recipient	VARCHAR(120)	
	content	TEXT	
	delivery_result	VARCHAR(24)	
	attempt_count	INTEGER	
	last_attempt_at	TIMESTAMP?	
	sent_at	TIMESTAMP?	
	failure_reason	TEXT?	

PK：主键；FK：外键；UK：唯一；UK*：联合唯一；?：可空。实线三角箭头：外键表 → 被引用表；同一对表的多个外键合并一条线，逐字段目标见表内。